

Rtools is a Windows-only toolchain.

On macOS there is no Rtools. Instead, R uses Apple's developer tools:

1. Install Xcode Command Line Tools:

⌘ Bash

```
xcode-select --install
```

This provides:

- `clang` (C/C++ compiler)
- `make`
- headers and libraries needed to compile R packages

2. Check that they are installed:

⌘ Bash

```
clang --version  
make --version
```

3. From within R:

⌘ R

```
system("clang --version")  
Sys.which("make")
```

For most packages, that's all you need.

If a package requires additional external libraries (e.g. `sf`, `terra`, `xml2`, `curl`, `openssl`), these are often installed via Homebrew:

⌘ Bash

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Then for example:

⌘ Bash

```
brew install gdal  
brew install geos  
brew install proj
```

The equivalent mapping is roughly:

Windows**macOS**

Rtools43

Xcode Command Line Tools

Rtools44

Xcode Command Line Tools

Rtools45

Xcode Command Line Tools

gcc/mingw

clang/LLVM

So if `renv::restore()` fails on Windows, checking Rtools is important. On macOS, the corresponding check is whether the Xcode Command Line Tools are installed and whether `clang` is available.